

# YouDescribe: Bridging AI Efficiency and Human Insight for Scalable Audio Description

Lana Do  
Northeastern University  
San Jose, USA  
do.ng@northeastern.edu

Sanjay Mirani  
San Francisco State University  
San Francisco, USA  
smirani1@mail.sfsu.edu

Charity Pitcher-Cooper  
Smith-Kettlewell Eye Research  
Institute  
San Francisco, USA  
cpc@ski.org

Xuan Duy Anh Nguyen  
San Francisco State University  
San Francisco, USA  
anguyen96@sfsu.edu

Alekya Bairaboina  
San Francisco State University  
San Francisco, USA  
abairaboina@sfsu.edu

Ilmi Yoon  
Northeastern University  
San Jose, USA  
i.yoon@northeastern.edu

## Abstract

Audio description (AD) is critical for making video content accessible to blind and low-vision (BLV) audiences, yet remains largely unavailable for short-form and user-generated videos. YouDescribe addresses this gap by enabling BLV users to request, rate, and contribute to descriptions, while sighted volunteers create or refine AD through a structured, interactive interface. The platform's latest version introduces drafts generated using Artificial Intelligence (AI) within a Human-in-the-Loop (HITL) workflow, helping describers accelerate the authoring process while preserving narrative accuracy and contextual nuance. Importantly, its accessible design ensures BLV users are not passive consumers, but active contributors—enhancing the inclusivity and relevance of the content. This paper presents YouDescribe's innovative design and system architecture, exploring how AI-assisted collaboration enhances the scalability, quality, and inclusivity of audio description.

## CCS Concepts

• **Human-centered computing** → **HCI**; • **Accessibility technologies**; • **Artificial intelligence**;

## Keywords

human-in-the-loop, multimodal AI system, audio description, blind and low-vision audiences

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## 1 Introduction

Digital video has become a dominant medium for communication, education, and entertainment, yet much of it remains inaccessible to blind and low-vision (BLV) users due to its reliance on non-verbal cues and the lack of accessibility features. Audio description (AD) provides spoken narration of key visual elements, enabling BLV users to better engage with video content. Research shows that AD significantly improves comprehension and enjoyment for visually impaired audiences [11, 16]. Despite its importance, AD remains largely unavailable—especially on short-form, user-generated platforms like YouTube, TikTok, and Instagram. Professional AD production is costly and time-consuming [13], and while legally mandated for television and streaming services under the 21st Century Communications and Video Accessibility Act, no such requirement exists for social media. As a result, subscription-based platforms like Netflix and Disney+ now offer thousands of audio-described titles, while content-sharing sites such as YouTube provide little to no AD support—significantly limiting BLV users' access to crucial information, cultural trends, and educational resources.

To address this gap, YouDescribe was launched in 2013, pioneering crowdsourced audio description by sighted volunteers upon request [21]. Unlike traditional AD workflows that rely on trained describers, voice actors, and professional editing tools [3], YouDescribe lowers the barrier for amateur contributors through a guided interface and built-in audio tools. This enabled a growing community of over 3,000 volunteers to produce more than 6,400 audio-described videos by 2022 [14]. The rapid growth of video content has outpaced manual efforts, leaving many requests unfulfilled and emphasizing the need to move beyond a volunteer-only model. AI-driven systems offer a promising solution, but current models still struggle with narrative flow, contextual nuance, and audio timing—making fully automated AD insufficient on its own. A Human-in-the-Loop (HITL) approach, in which AI-generated drafts are refined by human contributors, is essential for preserving quality and contextual relevance.[22].

In 2025, YouDescribe rolled out new features, including AI-generated drafts and an intuitive, well-designed interface to streamline the editing and refinement process. New tools for adjusting narration timing, selecting speech options, and viewing contribution transparency support both experienced and novice describers. This paper presents the design innovations behind the updated YouDescribe

platform, illustrating how AI-assisted tools empower the community to scale high-quality audio description for BLV users.

## 2 Background and Related Work

### 2.1 Volunteer-Driven AD Platforms

Volunteer-driven initiatives have played a key role in expanding access to audio description for BLV users. LiveDescribe was one of the first tools to support community-generated AD [3]. ViScene investigated the impact of feedback mechanisms on novice describers' output [12], while AccessibleAD facilitated direct collaboration between describers and BLV users [9]. Among these efforts, YouDescribe stands out as one of the most scalable and widely used solutions for crowdsourced audio description. First launched in 2013, YouDescribe enables sighted volunteers to produce AD tracks synchronized with YouTube videos through both a web-browser and an iOS app. Training modules, including step-by-step tutorials, and intuitive tools for recording and timing narration make AD creation accessible to volunteers without formal training. A key feature is the Wishlist, which allowed BLV users to request specific videos—aligning describers' efforts with community needs. Over time, YouDescribe has generated thousands of described videos and a rich dataset supporting research in accessibility, language modeling, and video analysis [14]. To address the limitations of relying solely on human effort, YouDescribe introduced an enhanced version in 2025 with AI integration and a redesigned interface for greater scalability and usability. This shift reflects broader trends in AD generation, where AI-powered systems are increasingly explored as a means to support more efficient accessibility solutions.

### 2.2 AI-Powered Audio Description Systems

Recent systems have explored AI-based approaches to scale AD creation. Wang et al. (2021) proposed a system analyzing multimodal video content for automatic AD [19], while ShortScribe, LLM-AD, and MMAD leverage vision-language models like BLIP-2, GPT-4V, and LLaMA2 to produce scene-aware descriptions [7, 18, 20]. Despite their potential, these systems often lack narrative coherence or contextual nuance, making human oversight essential. Interactive AD systems address this gap by combining automation with human input. DescribeNow prompts GPT-4V using professional guidelines to respond to user queries [5], while hybrid approaches like that of Ihorn et al. generate descriptions throughout the video via a system called NarrationBot and supplement them with on-demand responses using InfoBot [2]. While these systems incorporate human input, YouDescribe takes it a step further by adopting the HITL model, in which AI-generated drafts are directly refined by human contributors through smart editing tools. This process ensures that AI supports—rather than replaces—human judgment, while centering the BLV community in the description workflow.

## 3 System Overview

### 3.1 Navigation and Features

The YouDescribe platform ensures a seamless experience for both BLV users and volunteer describers. Figure 1 provides an overview of the website's navigation structure, illustrating how its design supports an enhanced user experience.

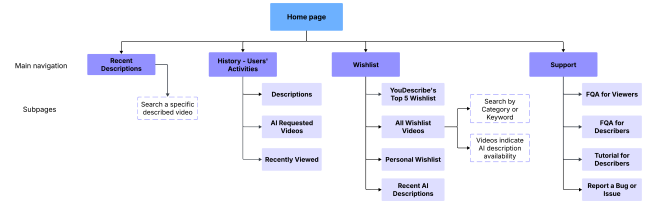


Figure 1: YouDescribe Homepage and Main Navigation

The **Recent Descriptions** section showcases newly available audio-described videos, whether added as freestyle descriptions or refined from an existing drafts. The search function for specific videos is accessible directly from the homepage. The **History** section helps users manage their activity by tracking viewed videos, monitoring AI-generated description requests, and revisiting completed or in-progress contributions. The **Wishlist** has long been a staple of the YouDescribe platform, allowing blind and low-vision (BLV) users to request descriptions for videos lacking AD and enabling describers to prioritize content based on community demand. In the new version, this feature has been expanded with the introduction of a Top 5 Wishlist section, which highlights the most requested videos across the platform. Users can also curate a personal wishlist to keep track of videos they are individually interested in having described. Each video clearly displays whether an AI-generated draft is available, helping describers decide between creating a new freestyle description or refining an existing draft. Additional filtering by category or keyword allows volunteers to engage with content that aligns with their interests and expertise. The **Support** section offers FAQs, tutorials, and a bug-reporting system, ensuring users can navigate platform features, troubleshoot issues, and learn best practices for creating high-quality audio description. These resources encourage broader participation, reinforcing YouDescribe's mission to improve digital accessibility.

### 3.2 User Experience and Contribution Model

YouDescribe offers a seamless experience for BLV users to browse, request, and select from multiple AD versions when available, while enabling describers to create new descriptions or refine existing drafts—supporting scalable and efficient accessibility. Figure 2 shows the interaction flow for both user groups. The platform supports ARIA-based screen readers, high-contrast mode, and full keyboard navigation to ensure accessibility across devices. Figure 6 (see Appendix A) shows the video interface as experienced by a user.

Published videos may include multiple AD versions from different describers, allowing viewers to choose their preferred narration, rate it, and provide feedback. The audio ducking feature lets users adjust AD volume relative to the original video, ensuring clarity without disrupting key sounds or dialogue. If no AD is available, the video can be added to the Wishlist (if not already listed), and an AI-generated draft may be requested. Once available, users receive an email notification and can view and rate the draft—ensuring access even in the absence of human-authored descriptions.

Describers contribute in various ways: by writing original scripts, requesting AI-generated drafts as starting points, or editing existing

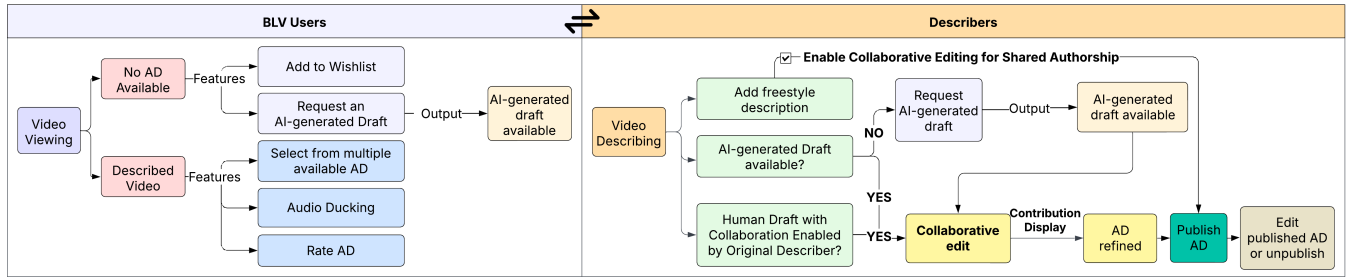


Figure 2: Video Interaction Flow for BLV Users and Describers

descriptions. Upon publishing, they may enable collaborative editing, allowing others to revise their version. This option is always enabled for AI-generated drafts to support the human-in-the-loop (HITL) model, but remains optional for human-authored content (see Figure 7, Appendix B). To promote transparency, YouDescribe displays the proportion of each describer’s contribution, including the ratio of AI-generated to human-edited content (see Figures 8 and 9, Appendix C). Contributors can revisit, update, or unpublish their work for further refinement.

While audio description has traditionally been a one-way process—produced by sighted volunteers and consumed by BLV audiences—YouDescribe encourages a more participatory approach to accessibility. With screen reader support and easy-to-navigate tools, BLV users are empowered to correct inaccuracies and improve AD directly through the editing interface. This marks a shift from passive viewer to active content creator. The result is a richer, more adaptive experience in which BLV individuals gain greater agency over the content they engage with, and describers receive meaningful feedback to better serve their intended audience. By bringing both groups together in a shared, collaborative space, YouDescribe supports a dynamic, community-driven model of accessibility that continues to evolve.

## 4 Audio Description Generation and Editing

### 4.1 Freestyle Description Interface

The YouDescribe freestyle interface provides a streamlined workflow for authoring original AD scripts. Descriptions are categorized as either Inline, which play during natural pauses, or Extended, which pause playback to allow for longer audio segments. While traditional AD typically relies on Inline delivery—especially in entertainment contexts like animations or films—YouDescribe’s Extended mode offers a unique advantage for educational or instructional content, enabling authors to explain complex visuals, steps, or diagrams in greater detail. Inline and Extended cues are visually marked in yellow and purple, respectively. Descriptions are further classified as either Visual (describing key visual elements) or Text on Screen (reading visible on-screen text). Authors can choose between synthesized speech using text-to-speech (TTS) or recording their own narration. TTS voices are differentiated: a female voice for Text on Screen and a male voice for Visual descriptions, helping BLV audiences distinguish between them.

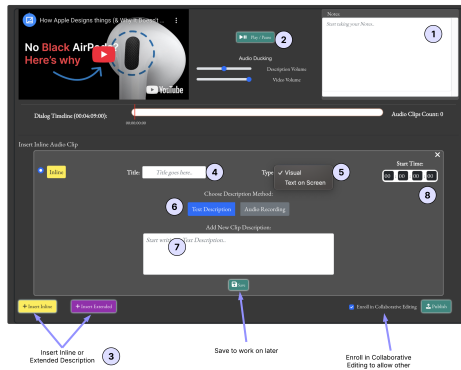
Figure 3 illustrates the description interface and outlines the process a describer follows to insert AD in freestyle mode. Once the

entire video has been described, the final result can be previewed with synchronized playback. Editors can adjust start times using the timestamp box (see bubble 8 in Figure 3) or fine-tune placement with the **nudging tool**, shifting cues left or right with precision. As the video plays, active descriptions are highlighted in green, making them easy to track in real time. Before publishing, authors can choose to enable collaborative editing. Progress is saved under My Descriptions, where described videos remain accessible for future edits or can be unpublished if needed. By combining a guided workflow with intuitive tools, YouDescribe makes the AD creation process more accessible, flexible, and efficient for contributors.

### 4.2 AI-Generated Description Pipeline

The AI-generated draft is produced through a GPU-accelerated pipeline, integrating computer vision and natural language processing techniques for efficient description generation [2]. Figure 5 outlines this process.

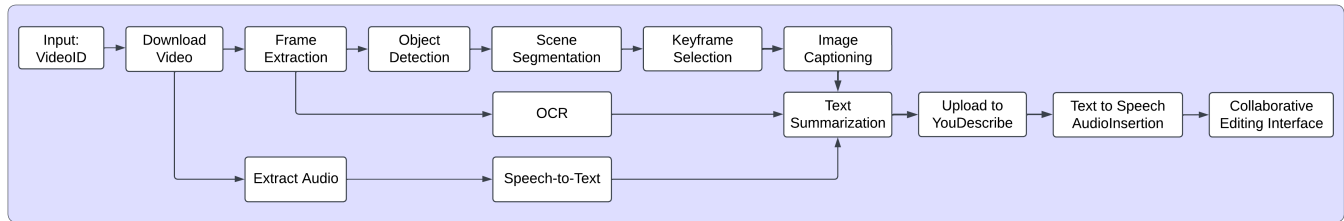
The pipeline begins with OpenCV, which captures still frames from the input video at regular intervals [1]. Next, FFmpeg—a multimedia framework—is used to isolate the audio track for audio transcription and alignment with visual information [17]. Optical Character Recognition (OCR) is then applied to detect on-screen text, while You Only Look Once version 3 (YOLOv3) performs real-time object detection to assist in scene segmentation [15]. Keyframes are identified by comparing each sampled frame with its predecessor using pixel-level differences. A frame is selected if over 30% of its pixels differ, indicating a significant visual change. These visually distinct frames are then processed using Bootstrapped Language-Image Pretraining version 2 (BLIP-2), a vision-language model that generates descriptive captions from images [10], and the audio is transcribed using Google Speech-to-Text. The resulting outputs—including recognized on-screen text, image captions for segmented scenes, and transcripts—are synthesized into one comprehensive and coherent description. The descriptions, categorized as Text on Screen or Visual, are uploaded to the YouDescribe server in JSON format. Google TTS converts the text into speech, which is synchronized with the video and inserted as either Inline or Extended descriptions based on audio length and available dialogue gaps. The finalized AD is then made available for volunteer refinement through collaborative editing.



**Figure 3: Freestyle AD creation interface. Workflow: (1) Take notes, (2) Play/Pause to insert AD, (3) Choose Inline/Extended, (4) Add title, (5) Select type, (6) Choose audio, (7) Write (8) Adjust timing.**



**Figure 4: Collaborative Editing AI-Generated AD Interface. Describers can refine text, adjust timing with the nudge tool, add new AD, or delete inaccurate ones.**



**Figure 5: GPU-accelerated pipeline for generating AI-based audio description for videos.**

### 4.3 Collaborative Editing of AI-Generated Descriptions

Once the GPU-accelerated pipeline generates AI descriptions, they are integrated into the collaborative editing interface. As shown in Figure 4, this interface closely resembles the freestyle description interface, providing a familiar environment for describers while streamlining the workflow with pre-populated, timestamped descriptions. The describer can see how many audio description are generated for the whole video. Similarly to freestyle editing mode, describers can preview the AI-generated draft with active AD highlighted in green.

Each AI-generated description is pre-filled and categorized as Visual or Text on Screen, with insertion labeled as Inline or Extended. Describers can edit text, adjust timestamps, switch insertion modes, nudge timing for precision, add new descriptions, or delete unnecessary ones. Additionally, they can replace AI-generated speech with recorded narration for greater expressiveness. After publishing, users can see the contribution transparency, displaying the proportion of AI-generated content versus human edits. By integrating AI assistance with human oversight, YouDescribe offers a scalable and high-quality approach to video accessibility. The collaborative editing mode makes the process less daunting by giving describers a initial AI-generated versions to refine, rather than requiring them to writing descriptions from the ground up. This significantly lowers the barrier for participation while maintaining accuracy, clarity,

and contextual depth in the final descriptions. For more detailed examples of these enhancements, refer to Figure 10 in Appendix D.

## 5 Conclusion and Future Works

YouDescribe improves video accessibility for BLV users by expanding the availability of audio description, particularly for short-form and user-generated content that traditionally lacks accessibility features. By integrating AI-generated drafts, the new version of the platform enables BLV users to access descriptions more quickly, with the ability to request, review, and rate AI-generated AD, enhancing their control over the experience. For describers, YouDescribe reduces the burden of creating descriptions from scratch by providing generated AI-assisted outlines that optimize the process. The collaborative editing interface fosters a community-driven approach, improving AD quality through iterative human refinement while maintaining transparency in AI-versus-human contributions. These features not only improve the efficiency and scale of AD production but also shift traditional roles—encouraging BLV users to move from passive consumers to active participants in the creation process. By centering accessibility, co-creation, and thoughtful design, YouDescribe lays the foundation for more inclusive and adaptive audio description systems.

While YouDescribe has improved video accessibility, challenges remain in refining AI-generated descriptions to meet diverse user needs and align with AD guidelines. Current models struggle with

contextual reasoning, particularly across video genres and user preferences. Research shows that BLV users vary in their AD needs—for example, preferring detailed narration for social videos and concise AD for educational content [6, 8]. Immersive AD studies also emphasize the value of customizable experiences, such as interactive object exploration in 360° video [4]. Future work will focus on improving AI-generated AD quality using advanced Vision-Language Models (VLMs) for better scene understanding and natural sounding audio description. YouDescribe also plans to support user-driven customization through features like Infobot [2], allowing BLV users to pause and prompt questions. These efforts aim to deliver more personalized and high-quality AD experiences while advancing AI-human collaboration.

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A User Interface for Video with Different AD Options

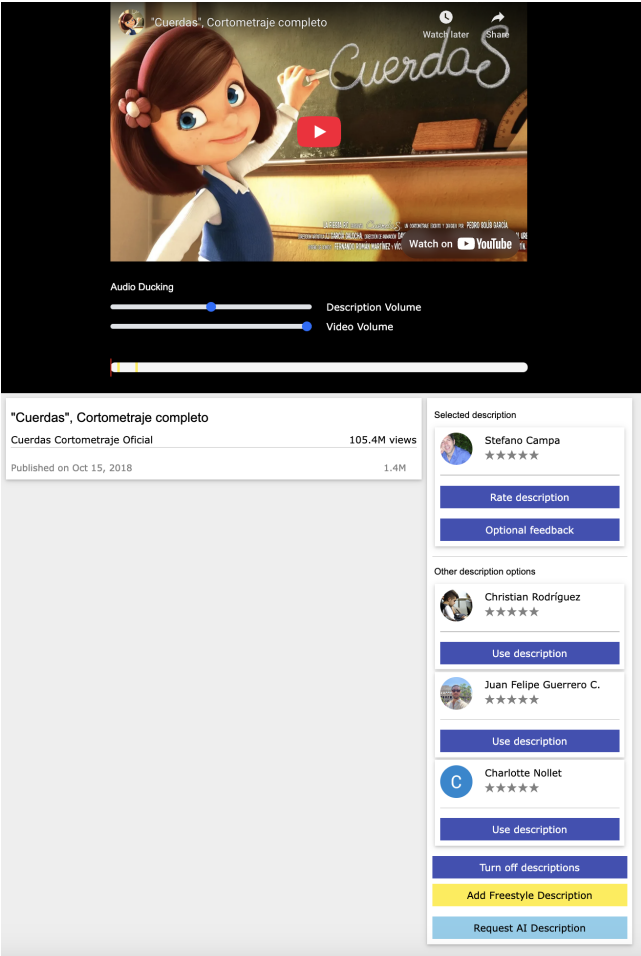
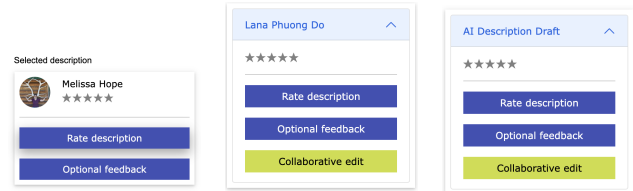


Figure 6: User Interface for Video with Different AD Options

B Editing Permissions and Collaboration



- (a) Describer opts out of collaborative edit; only their input is recorded
- (b) Describer enables collaborative edit, allowing others to contribute
- (c) Collaborative edit is enabled by default for AI drafts

Figure 7: Three different modes of collaborative editing

C Audio Description Contribution Breakdown

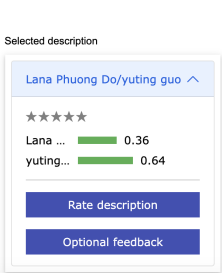


Figure 8: Contribution breakdown from multiple describers



Figure 9: Contributions of describers and AI-generated draft

D Examples of Human-in-the-Loop (HITL) Improvements to Audio Description

HITL editing allows volunteers to refine AI-generated audio description by correcting inaccuracies, improving fluency, and enhancing clarity. These human edits are essential for ensuring that descriptions are accurate, contextually appropriate, and accessible for BLV users. The examples below illustrate common improvements made during this process, ranging from factual corrections and stylistic enhancements to the addition of critical information and simplification for better pacing.



AI-generated description: A man in a green jacket walks through a park, when another person holding a netted object places it on him. The person then takes out a lighter from his pocket and pokes the man in the green jacket with it.

Human refined description: **Mr. Bean traps the camera thief** using a netted trash can. He then pulls a pen from his pocket and pokes the thief with it.

(a) Video titled *Camera Thief* |Mr. Bean Official Cartoon. Human refinement corrects inaccuracies in the AI-generated description.



AI-generated description: A young man, visibly distressed, kneels beside an older man who lies on the ground, appears injured on the street.

Human refined description: **Peter**, visibly distressed, kneels beside **Uncle Ben**, offering comfort as he lies injured on the street following the carjacking.

(b) Video titled *Uncle Ben's Death Vengeance Scene - Spider-Man (2002)*. Human revision adds context and improves the naturalness of the AI-generated description.



AI-generated description: A person opens a can of tuna, drains the oil, and adds chopped onion, pickles, celery, mayonnaise, crushed red pepper, lemon juice, black pepper into a bowl. The ingredients are mixed together to create a tuna salad.

Human refined description: A person opens a can of tuna about 150 grams, drains the oil, and adds 35 grams chopped onion, 35 grams chopped pickles, 35 grams chopped celery, 60 grams of mayonnaise, 0.7 table spoon of crushed red pepper, 1 table spoon of lemon juice, 0.5 teaspoon of black pepper into a bowl. The ingredients are mixed together to create a tuna salad.

(c) Video titled *Tuna Melt Sandwich :: Luch Box*. Human edit enhances detail by adding precise measurements in a cooking scene.



AI-generated description (**Extended**): The video showcases a lush, green forested landscape with dense foliage and rolling hills under a partly cloudy sky.

Human refined description (**Inline**): A lush forest with rolling hills appears under a partly cloudy sky.

AI-generated description (**Extended**): The final scene captures a chimpanzee perched in the treetops, moving gracefully among the branches.

Human refined description (**Inline**): A chimpanzee moves gracefully through the treetops

(d) Video titled *jane goodall*. Human edit condenses an overly long description to improve pacing and clarity.

Figure 10: Examples of human-in-the-loop improvements that enhance the quality, accuracy, and accessibility of AI-generated audio description.